

6.3 Permutations and Combinations

Thursday, November 7, 2024 6:25 PM

How many ways can we arrange 4 different cubes in a row?

$$\overline{4} \cdot \overline{3} \cdot \overline{2} \cdot \overline{1} \rightarrow 4!$$

What if we start with 7 different cubes?

How many ways can we arrange 4 of them in a row?

$$\overline{7} \cdot \overline{6} \cdot \overline{5} \cdot \overline{4} \rightarrow 7 \cdot 6 \cdot 5 \cdot 4$$

Definition: An ordered arrangement of a set of r distinct objects is an r -permutation.

Let $S = \{a, b, c\}$ a 2-perm of S is (a, c) .

Another 2-perm is (c, a) . These are different since they are in a different order.

Theorem: If $n, r \in \mathbb{Z}^+$ with $1 \leq r \leq n$, then the number of r -permutations of a set with n distinct elements is

$$P(n, r) = n(n-1)(n-2)\dots(n-r+1)$$

May also see this written P_r^n .

Theorem: If $n, r \in \mathbb{Z}^+$ with $1 \leq r \leq n$, then the number of r -permutations of a set with n distinct elements is

$$P(n, r) = \frac{n!}{(n-r)!}$$

How many permutations of the letters ABCDEFGH contain the string "ABC".

$$\begin{array}{cccccc} \overline{6} & \overline{5} & \overline{4} & \overline{3} & \overline{2} & \overline{1} \\ \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow \\ \text{"ABC"} & & & & & \\ D & & & & & \\ E & & & & & \\ F & & & & & \\ G & & & & & \end{array} \rightarrow 6! = P(6, 6)$$

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Definition: An unordered arrangement of r distinct elements is an r -combination.

An unordered arrangement is just a set.

If I have 7 different cubes, how many different groups of 4 cubes can I make?

$\overline{7} \quad \overline{6} \quad \overline{5} \quad \overline{4}$ 7·6·5·4 different
 $\downarrow \quad \downarrow \quad \downarrow \quad \downarrow$ arrangements of 4
 7 · 6 · 5 · 4 cubes.

$\overline{4} \quad \overline{3} \quad \overline{2} \quad \overline{1}$ 4! different arrangements
 4 · 3 · 2 · 1 of 4 cubes.

$\Rightarrow \frac{7 \cdot 6 \cdot 5 \cdot 4}{4 \cdot 3 \cdot 2 \cdot 1}$ different groups of 4.

Theorem: The number of r -combinations of a set with n elements (where $0 \leq r \leq n$) is

$$C(n, r) = \frac{n!}{r!(n-r)!} = \binom{n}{r}$$

Also written C_r^n

$\binom{n}{r}$ is a binomial coefficient read "n choose r."

Here's another way to see the relation between $P(n, k)$ and $C(n, k)$

If I want to count the number of k -combinations from n objects then we still start the same way as with k -permutations.

$\overbrace{\quad \quad \quad \dots \quad}^k$
 $\uparrow \quad \uparrow \quad \uparrow \quad \dots \quad \uparrow$
 $n \quad n-1 \quad n-2 \quad \dots \quad n-(k-1)$ $n(n-1)(n-2) \dots (n-k+1)$

but this is counting every rearrangement of each combination.

So how many rearrangements are there for each group of k objects?

$$\begin{array}{ccccccc} \text{---} & \text{---} & \text{---} & \cdots & \text{---} & \text{---} & \\ \uparrow & \uparrow & \uparrow & \cdots & & \uparrow & \\ k & k-1 & k-2 & \cdots & & 1 & \end{array} \quad k(k-1)(k-2)\cdots 2 \cdot 1$$

So each group is being "double" counted by a factor of $k!$

Then if we divide by $k!$, we remove all the repeated groups and get the desired count.

$$\frac{n(n-1)(n-2)\cdots(n-k+1)}{k(k-1)\cdots 2 \cdot 1} = \frac{n!}{(n-k)!} \cdot \frac{1}{k!} = \frac{n!}{(n-k)!k!}$$

How many 5-card hands can be dealt from a standard 52 card deck?

Order of cards in your hand doesn't matter so this is asking for the number of 5-combinations from a set of 52 objects.

$$C(52, 5) = \binom{52}{5} = \frac{52!}{5!(47)!}$$

Is $C(n, k) = C(n, n-k)$?

Explain symbolically.

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$$C(n, n-k) = \frac{n!}{(n-k)!(n-(n-k))!}$$
$$= \frac{n!}{(n-k)!(k)!} = C(n, k)$$

How many injective functions are there with domain $\{1, 2, 3\}$ and codomain $\{a, b, c, d, e, f\}$?

An injective function uses exactly 3 distinct elements of the codomain

$$f(1) = \square \leftarrow 6 \text{ choices}$$

$$f(2) = \square \leftarrow 5 \text{ choices}$$

$$f(3) = \square \leftarrow 4 \text{ choices}$$

so there are $6 \cdot 5 \cdot 4$ many functions.

We looked for arrangements of 3 objects ($k=3$). Order matters so this is an ordered arrangement i.e. 3-permutations.

So there are $P(6, 3)$ many functions
Cardinality of codomain

How many bit strings of length 10 are there with exactly 2 zeros?

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out of 10 positions, we want to find all possible ways to choose 2 positions to fill with a 0.

Can also think of it as all the ways to choose two distinct index values of a length 10 array.

Then we want an arrangement of 2 values (2 positions/2 index values)

Does the order we get these positions matter?

No, they both get set to zero.

Then we want all 2-combinations from a set of 10

$$\text{i.e. } C(10, 2) = \binom{10}{2} = \frac{10!}{2!8!}$$

How many bit strings of length 10 contain at least 8 ones?

To get at least 8 we could have exactly 8, exactly 9, exactly 10 ones.

No overlap. Using last example and the sum rule we get

$$\begin{aligned} & C(10, 8) + C(10, 9) + C(10, 10) \\ &= \binom{10}{8} + \binom{10}{9} + \binom{10}{10} \quad \text{many} \end{aligned}$$

How many permutations of the letters A B C D E F G contain the string "BCD" exactly?

I want arrangements where order matters, but BCD can't change.

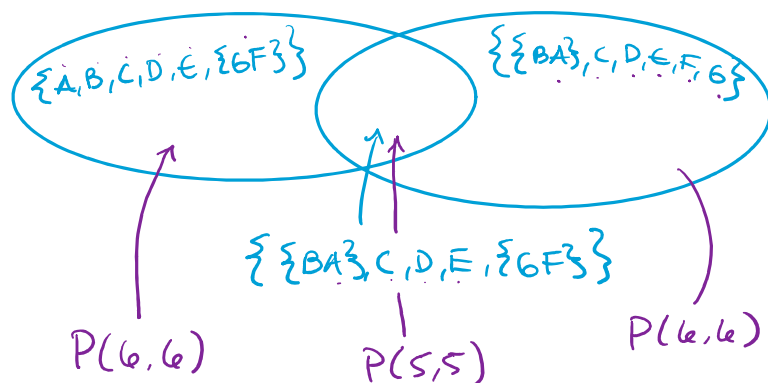
This is the same as arrangements of the set $\{A, \{BCD\}, E, F, G\}$

of the set $\{A, \{BCD\}, E, F, G\}$

So we want all 5-permutations of this set!

There are $P(5,5)$ many such permutations of the letters

How many permutations of the same letters contain the strings "BA" or "GF" exactly.



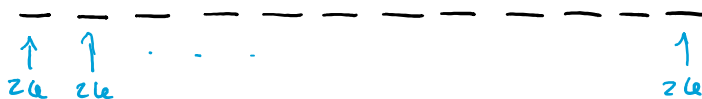
By the Inclusion-Exclusion Principle

There are $P(6,6) + P(6,6) - P(5,5)$
 $= 2P(6,6) - P(5,5)$ many

How many 12-character passwords are there which use a-z and 0-9 only if you can have at most four digits which must appear as consecutive characters?

0 digits, 1 digit, 2 digits, 3 digits, 4 digits
are all cases.

○ digits $\Rightarrow (26)^{12}$ many



1 digit $\Rightarrow$ 11 letters

_ # _ # _ # _ # _ # _ # _ # _ # _

the digit can appear in any 1 of the above 12 positions $\Rightarrow C(12,1)$

Then choose the digit and choose letters.

$$\begin{aligned} \text{so there are } & \binom{\# \text{ digit positions}}{\binom{\text{digit choices}}{\binom{\text{letter choices}}}} \\ & = C(12,1) \cdot 10 \cdot 26^{11} \text{ many.} \end{aligned}$$

2 digits $\Rightarrow$ 10 letters

_ # _ # _ # _ # _ # _ # _ # _

Same comment as above but now

represents two digits.

$$\begin{aligned} \text{there are } & \binom{\# \text{ digit positions}}{\binom{\text{digit choices}}{\binom{\text{letter choices}}}} \\ & = C(11,2) \cdot 10^2 \cdot 26^{10} \text{ many} \end{aligned}$$

Following this procedure and applying the sum rule leaves us with

$$\begin{aligned} & 26^{12} + C(12,1) \cdot 10 \cdot 26^{11} + C(11,2) \cdot 10^2 \cdot 26^{10} \\ & + C(10,3) \cdot 10^3 \cdot 26^9 + C(9,4) \cdot 10^4 \cdot 26^8 \\ & \text{many such passwords.} \end{aligned}$$

(This is roughly 7.64×10^{17})